

① Tabelas-verdade e expressão booleanas

1.

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

c \ AB	00	01	11	10
0	0	0	1	0
1	1	1	1	1

$$F = C + AB$$

2.

A	B	C	G
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

c \ Ab	00	01	11	10
0	0	1	0	0
1	1	1	1	1

$$G = C + \bar{A}B$$

② Simplificação com álgebra booleana

$$\begin{aligned} \textcircled{6} F &= AB + \cancel{A\bar{B}} + \cancel{\bar{A}B} \\ &= AB \end{aligned}$$

$$\begin{aligned} \textcircled{10} F &= A + \bar{A}B \\ &= (A + \bar{A})(A + B) \\ &= A + B \end{aligned}$$

$$\begin{aligned} \textcircled{7} F &= (A + \bar{B})(A + \bar{B}) \\ &= A \end{aligned}$$

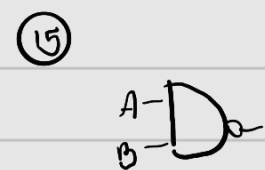
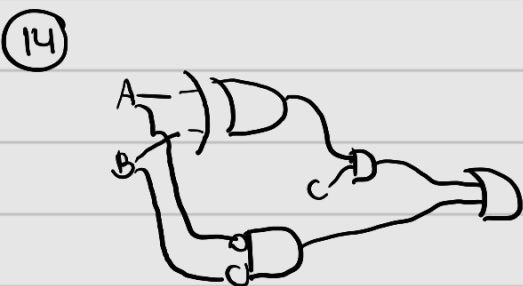
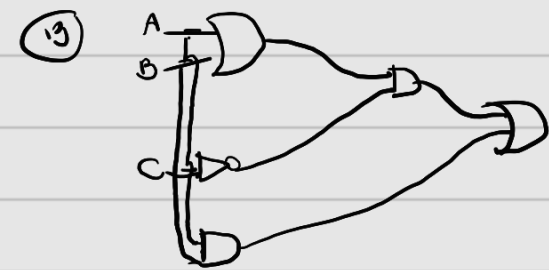
$$\begin{aligned} \textcircled{11} F &= (A + B + C)(A + \bar{B} + C) \\ &= (A + A)(A + \bar{B})(A + C)(\bar{B} + A)(\bar{B} + \bar{B})(\bar{B} + C)(C + A)(C + \bar{B})(C + C) \\ &= A + C \end{aligned}$$

$$\begin{aligned} \textcircled{8} F &= \cancel{A\bar{B}C} + \cancel{A\bar{B}C} + \cancel{AC\bar{B}} + \cancel{AC\bar{B}} \\ &= AC \end{aligned}$$

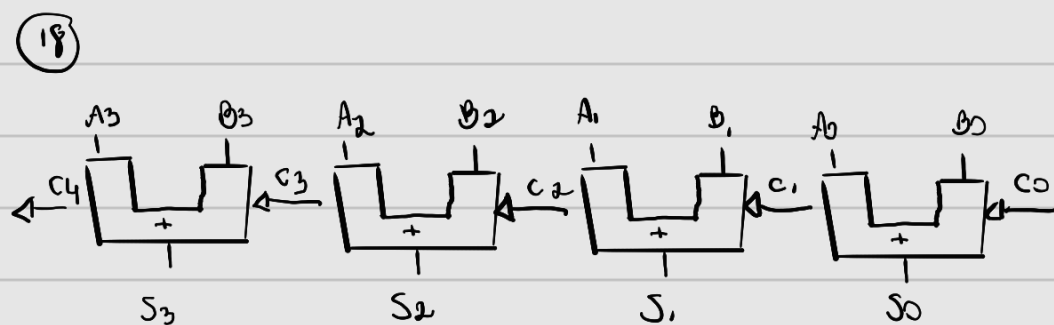
$$\begin{aligned} \textcircled{12} F &= \bar{A}B + \bar{A}C + BC \\ &= \bar{A}B + AC \end{aligned}$$

$$\begin{aligned} \textcircled{9} F &= \cancel{\bar{A}\bar{B}} + \cancel{\bar{A}\bar{B}} + \bar{A}\bar{B}C \\ &= \bar{A} + \bar{B}C \end{aligned}$$

③ Expressões para circuitos



④ Sumadores



5) circuitos para expressões

$$\textcircled{20} Y = A\bar{B} + AB + AC \\ = A$$

$$\textcircled{21} G = A\bar{C} + BC + AB \\ = A + BC$$

$$\textcircled{22} X = A \oplus B$$